Project Design Phase-II
Technology Stack (Architecture & Stack)

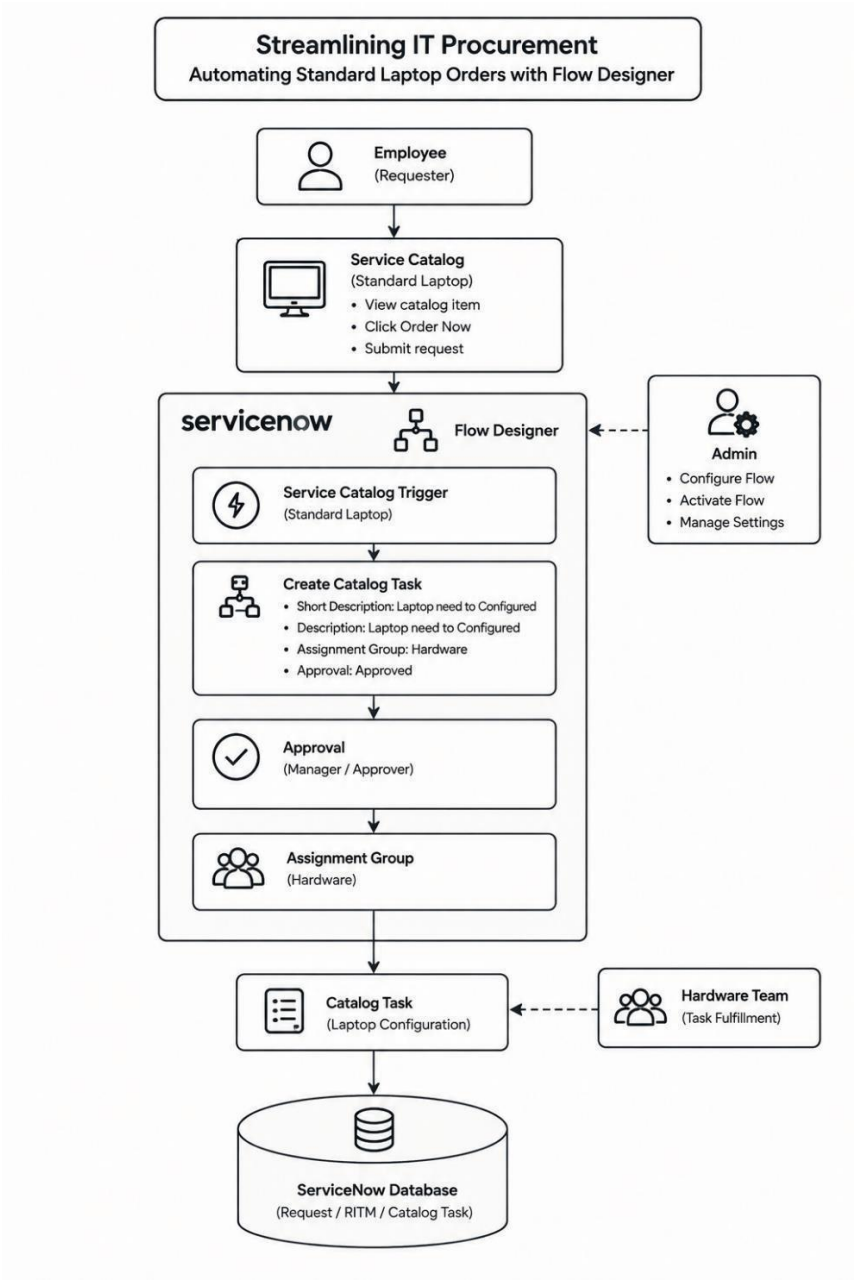
Date	3 October 2026
Team ID	SWTID-2026-2050
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Technical Architecture:

The system automates standard laptop procurement using ServiceNow Service Catalog and Flow Designer. An employee submits the Standard Laptop request, the Service Catalog creates the request and Requested Item, and the Service Catalog trigger starts the Standard Laptop Task flow. The flow automatically creates a Catalog Task and assigns it to the Hardware group for fulfillment.

Example: Streamlining IT Procurement – Automating Standard Laptop Orders with Flow Designer

Reference: ServiceNow Flow Designer and Service Catalog concepts used for implementing automated request workflows.



Architecture Requirement Implementation in Our Project

ServiceNow Service Catalog is used for submitting **Application Logic /** the Standard Laptop request. Flow Designer contains **Technology Blocks** the Service Catalog trigger and **Create Catalog Task** action.

The application is implemented and executed on **Infrastructure –** the **ServiceNow cloud platform**. No local **Local / Cloud** application server is required.

No external or third-party API is used in the **External Interfaces /** implemented Standard Laptop procurement **ThirdParty APIs** workflow.

ServiceNow stores the Request, Requested Item (RITM), **Data** Catalog Task, approval and related workflow information in **Storage** the ServiceNow platform database.

Machine Learning Not applicable. The project does not use a **Model** machine-learning model.

User Employees use the **ServiceNow Service Catalog** to access **Interface** and order the Standard Laptop catalog item.

ServiceNow Flow Designer automatically processes **Workflow** the submitted request and creates the required **Automation** Catalog Task.

Task The generated Catalog Task is assigned to the **Hardware** **Assignment** assignment group.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees interact with the system through the Service Catalog to select and order a Standard Laptop.	ServiceNow Service Catalog.
2.	Application Logic-1	Receives the Standard Laptop request and starts the automated procurement workflow.	ServiceNow Flow Designer – Service Catalog Trigger
3.	Application Logic-2	Automatically creates a catalog task for the requested laptop.	Flow Designer – Create Catalog Task Action
4.	Application Logic-3	Handles approval and routes the request/task to the appropriate assignment group.	ServiceNow Approval & Assignment Group – Hardware
5.	Database	Stores request, Requested Item (RITM), catalog task and workflow-related records.	ServiceNow Platform Database
6.	Cloud Database	ServiceNow manages platform data storage as part of its cloud platform.	ServiceNow Cloud Database

7.	File Storage	No dedicated file-storage component is required for the standard laptop procurement workflow.	Not Applicable
8.	External API-1	No external API is required for the standard laptop request and catalog task automation.	Not Applicable
9.	External API-2	No second external API is used in the implemented workflow.	Not Applicable
10.	Machine Learning Model	The project does not use a machine-learning model.	Not Applicable
11.	Infrastructure (Server / Cloud)	The application and workflow are executed on the ServiceNow cloud platform.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source framework is required. The application is developed using ServiceNow platform capabilities.	ServiceNow Flow Designer, Service Catalog

2.	Security Implementations	Access to the ServiceNow application and records is controlled through user authentication, roles and permissions. The workflow runs using the configured **System User**.	ServiceNow Authentication, Roles & Access Controls
3.	Scalable Architecture	The workflow is designed as a cloud-based automation process. Additional catalog requests can be processed using the same standardized flow without creating a separate workflow for each request..	ServiceNow Cloud Platform, Flow Designer
4.	Availability	The application is hosted on the ServiceNow cloud platform, allowing users to access the Service Catalog and submit requests through the platform.	ServiceNow Cloud Infrastructure
5.	Performance	Flow Designer automates catalog task creation and reduces manual processing. Performance can be evaluated by checking flow execution, task creation time and successful processing of multiple requests.	ServiceNow Flow Designer

References:

1. **C4 Model – Software Architecture Diagramming https://c4model.com/**
   Used as a reference for representing the system architecture using clear components, relationships, and levels of abstraction.
2. **ServiceNow – Create a Flow with a Service Catalog Trigger**
   ServiceNow Flow Designer – Service Catalog Trigger
   Supports the implementation of a flow that starts when the Standard Laptop Service Catalog item is requested.

3. **ServiceNow – Create Catalog Task Action**
   ServiceNow Create Catalog Task Action
   Supports the use of Create Catalog Task to create a catalog task associated with the Requested Item.
4. **IBM Cloud Architecture IBM Cloud Architecture**
   General reference for documenting cloud-based technical architecture.
5. **AWS Architecture AWS Architecture**
   General reference for architectural design and cloud infrastructure concepts.